

TASK 3 — SQL FOR DATA ANALYSIS

E-commerce Sales Analytics • Top-Performer Submission

Objective: Extract, transform and analyze structured e-commerce data using SQL, while demonstrating practical business thinking, joins, aggregation, subqueries, CTEs, window functions, views and query optimization.

Dataset	ecommerce_sales_analytics_5000.csv
Rows / columns	5,000 / 12
SQL engine	SQLite-compatible SQL
Analysis focus	Revenue • Customers • Categories • Regions • Payments • Delivery • Ratings
Deliverables	SQL queries + executed outputs + insights + optimization + interview preparation

Analyst signature: A clean, business-oriented SQL workflow designed to show not only how to write queries, but how to turn query results into decisions.

1. Executive Snapshot

Metric	Result
Total revenue	■5,109,775.74
Orders	5,000
Unique customers	989
Average order revenue	■1,021.96
Average customer rating	2.97 / 5
Average delivery time	6.12 days
Top category	Electronics — ■1,829,899.22
Top region	West — ■1,345,582.16

Top-performer angle: The report goes beyond basic SELECT statements. It uses SQL to answer management questions: Where does revenue come from? Which customers matter most? Which payment methods dominate? Does delivery experience align with ratings? Which categories deserve attention?

2. Dataset & Data Model

The supplied dataset contains transactional e-commerce records. The working SQL model keeps the original **sales** table and adds lightweight customer and product reference tables so that INNER JOIN and LEFT JOIN can be demonstrated realistically without changing the original transaction data.

order_id	order_date	customer_id	product_category	region	quantity	unit_price
10001	2022-01-01 00:00:00	1102	Beauty	South	7	373.65
10002	2022-01-02 00:00:00	1435	Clothing	South	7	47.74
10003	2022-01-03 00:00:00	1860	Beauty	East	3	311.28
10004	2022-01-04 00:00:00	1270	Electronics	West	5	524.47
10005	2022-01-05 00:00:00	1106	Clothing	West	5	139.87
10006	2022-01-06 00:00:00	1071	Electronics	West	2	264.44
10007	2022-01-07 00:00:00	1700	Clothing	South	6	564.60
10008	2022-01-08 00:00:00	1020	Beauty	North	6	265.71

Core fields: order_id, order_date, customer_id, product_category, region, quantity, unit_price, discount, payment_method, delivery_days, customer_rating and revenue.

3. Core SQL Analysis

01. Basic SELECT + filtering

```
SELECT order_id, order_date, product_category, region, revenue
FROM sales
WHERE revenue > 1500
ORDER BY revenue DESC
LIMIT 10;
```

Executed output:

order_id	order_date	product_category	region	revenue
10810	2024-03-20	Electronics	South	4,119.33
14884	2035-05-16	Electronics	South	4,083.17
12215	2028-01-24	Electronics	South	4,055.29
13102	2030-06-29	Beauty	East	4,005.93
13687	2032-02-04	Clothing	North	3,928.63
10158	2022-06-07	Electronics	North	3,917.54
13350	2031-03-04	Electronics	West	3,898.48
10322	2022-11-18	Electronics	West	3,893.11

02. Regional revenue

```
SELECT region,
COUNT(*) AS orders,
SUM(quantity) AS units_sold,
ROUND(SUM(revenue),2) AS total_revenue,
ROUND(AVG(revenue),2) AS avg_order_revenue
FROM sales
GROUP BY region
ORDER BY total_revenue DESC;
```

Executed output:

region	orders	units_sold	total_revenue	avg_order_revenue
West	1289	5260	1,345,582.16	1,043.90
North	1276	5164	1,281,508.45	1,004.32
South	1208	4884	1,246,640.90	1,031.99
East	1227	4916	1,236,044.23	1,007.37

03. Category performance

```
SELECT product_category,
COUNT(*) AS orders,
SUM(quantity) AS units_sold,
ROUND(SUM(revenue),2) AS revenue,
ROUND(AVG(customer_rating),2) AS avg_rating
FROM sales
GROUP BY product_category
ORDER BY revenue DESC;
```

Executed output:

product_category	orders	units_sold	revenue	avg_rating
Electronics	1777	7109	1,829,899.22	2.95
Clothing	1531	6171	1,531,931.72	3.01
Home	969	3949	982,083.92	2.94
Beauty	723	2995	765,860.88	3.01

04. Payment method mix

```
SELECT payment_method,
COUNT(*) AS orders,
ROUND(100.0*COUNT(*)/(SELECT COUNT(*) FROM sales),2) AS order_share_pct,
ROUND(SUM(revenue),2) AS revenue
FROM sales
GROUP BY payment_method
ORDER BY revenue DESC;
```

Executed output:

payment_method	orders	order_share_pct	revenue
Card	2270	45.40	2,366,248.20
COD	1774	35.48	1,788,408.44
Wallet	956	19.12	955,119.10

05. Monthly revenue trend

```
SELECT strftime('%Y-%m', order_date) AS month,
COUNT(*) AS orders,
ROUND(SUM(revenue),2) AS revenue
FROM sales
GROUP BY month
ORDER BY month;
```

Executed output:

month	orders	revenue
2022-01	31	35,624.89
2022-02	28	29,908.89
2022-03	31	30,309.28
2022-04	30	30,781.96
2022-05	31	41,629.24
2022-06	30	29,847.33
2022-07	31	28,686.24
2022-08	31	29,186.45

06. HAVING: high-value regions

```
SELECT region,
ROUND(SUM(revenue),2) AS revenue,
ROUND(AVG(customer_rating),2) AS avg_rating
FROM sales
GROUP BY region
HAVING SUM(revenue) > (SELECT AVG(region_revenue)
FROM (SELECT SUM(revenue) AS region_revenue
FROM sales GROUP BY region))
ORDER BY revenue DESC;
```

Executed output:

region	revenue	avg_rating
West	1,345,582.16	2.96
North	1,281,508.45	2.99

07. Average revenue per user

```
SELECT ROUND(AVG(customer_revenue),2) AS avg_revenue_per_customer
FROM (
SELECT customer_id, SUM(revenue) AS customer_revenue
FROM sales
GROUP BY customer_id
) t;
```

Executed output:

avg_revenue_per_customer
5,166.61

08. Customer leaderboard

```
SELECT customer_id,
COUNT(*) AS orders,
ROUND(SUM(revenue),2) AS lifetime_revenue,
ROUND(AVG(customer_rating),2) AS avg_rating
FROM sales
GROUP BY customer_id
ORDER BY lifetime_revenue DESC
LIMIT 10;
```

Executed output:

customer_id	orders	lifetime_revenue	avg_rating
1,663.00	14.00	17,680.69	2.75
1,955.00	10.00	16,265.15	3.49
1,675.00	11.00	15,036.46	2.98
1,276.00	11.00	15,023.71	3.04
1,647.00	11.00	14,894.97	3.03
1,804.00	10.00	14,330.98	3.04
1,193.00	7.00	14,151.65	3.27
1,836.00	9.00	14,110.78	3.31

4. Joins, Advanced SQL & Business Analysis

09. INNER JOIN: sales + customer segment

```
SELECT c.customer_segment,
COUNT(s.order_id) AS orders,
ROUND(SUM(s.revenue),2) AS revenue
FROM sales s
INNER JOIN customers c ON s.customer_id = c.customer_id
GROUP BY c.customer_segment
ORDER BY revenue DESC;
```

customer_segment	orders	revenue
Value	1696	1,732,764.15
Premium	1659	1,728,167.10
Standard	1645	1,648,844.49

10. LEFT JOIN: category + margin reference

```
SELECT p.product_category,
p.category_margin,
COUNT(s.order_id) AS orders,
ROUND(SUM(s.revenue),2) AS revenue,
ROUND(SUM(s.revenue)*p.category_margin,2) AS est_gross_profit
FROM products p
LEFT JOIN sales s ON p.product_category = s.product_category
GROUP BY p.product_category, p.category_margin
ORDER BY est_gross_profit DESC;
```

product_category	category_margin	orders	revenue	est_gross_profit
Clothing	0.29	1531	1,531,931.72	444,260.20
Electronics	0.22	1777	1,829,899.22	402,577.83
Home	0.27	969	982,083.92	265,162.66
Beauty	0.34	723	765,860.88	260,392.70

11. Subquery: above-average orders

```
SELECT order_id, customer_id, product_category, revenue
FROM sales
WHERE revenue > (SELECT AVG(revenue) FROM sales)
ORDER BY revenue DESC
LIMIT 10;
```

order_id	customer_id	product_category	revenue
10810	1890	Electronics	4,119.33
14884	1146	Electronics	4,083.17
12215	1113	Electronics	4,055.29
13102	1699	Beauty	4,005.93
13687	1831	Clothing	3,928.63
10158	1779	Electronics	3,917.54
13350	1082	Electronics	3,898.48
10322	1608	Electronics	3,893.11
10662	1550	Clothing	3,868.11
14306	1283	Beauty	3,807.92

12. CTE: delivery experience

```
WITH delivery AS (
SELECT region,
COUNT(*) AS orders,
ROUND(AVG(delivery_days),2) AS avg_delivery_days,
ROUND(AVG(customer_rating),2) AS avg_rating
FROM sales
GROUP BY region
)
SELECT * FROM delivery
ORDER BY avg_delivery_days ASC;
```

region	orders	avg_delivery_days	avg_rating
West	1289	6.08	2.96
East	1227	6.10	2.97
North	1276	6.12	2.99
South	1208	6.17	2.97

13. Window function: category revenue share

```
WITH cat AS (
  SELECT product_category, SUM(revenue) AS revenue
  FROM sales GROUP BY product_category
)
SELECT product_category,
  ROUND(revenue,2) AS revenue,
  ROUND(100.0*revenue/SUM(revenue) OVER (),2) AS revenue_share_pct
FROM cat
ORDER BY revenue DESC;
```

product_category	revenue	revenue_share_pct
Electronics	1,829,899.22	35.81
Clothing	1,531,931.72	29.98
Home	982,083.92	19.22
Beauty	765,860.88	14.99

14. Ranking: top 3 categories by region

```
WITH x AS (
  SELECT region, product_category,
  SUM(revenue) AS revenue
  FROM sales
  GROUP BY region, product_category
),
ranked AS (
  SELECT *,
  DENSE_RANK() OVER (PARTITION BY region ORDER BY revenue DESC) AS rnk
  FROM x
)
SELECT region, product_category, ROUND(revenue,2) AS revenue, rnk
FROM ranked
WHERE rnk <= 3
ORDER BY region, rnk;
```

region	product_category	revenue	rnk
East	Electronics	411,803.35	1
East	Clothing	402,582.60	2
East	Home	246,749.17	3
North	Electronics	472,450.93	1
North	Clothing	364,659.43	2
North	Home	258,184.77	3
South	Electronics	487,055.68	1
South	Clothing	348,363.14	2
South	Home	220,148.10	3
West	Electronics	458,589.26	1

5. Data Quality & Optimization

A strong analyst validates the data before trusting the dashboard or recommendation. The audit query checks missing identifiers, missing revenue, negative revenue and uniqueness of order IDs.

null_order_id	null_customer_id	null_revenue	negative_revenue	unique_orders
0	0	0	0	5000

Indexes used for optimization:

```
CREATE INDEX idx_sales_customer ON sales(customer_id);  
CREATE INDEX idx_sales_region_date ON sales(region, order_date);  
CREATE INDEX idx_sales_category ON sales(product_category);
```

Indexing improves lookup and filtering performance when columns are frequently used in WHERE clauses, JOIN conditions or grouping patterns. Indexes should be chosen based on workload and tested with an execution plan; unnecessary indexes add storage and write overhead.

Optimization checklist

1. Select only required columns.
2. Filter early.
3. Index high-use join/filter columns.
4. Avoid functions on indexed columns when possible.
5. Prefer appropriate joins and aggregations.
6. Use EXPLAIN / query plans to verify the real bottleneck.
7. Aggregate at the correct grain before joining large tables.

6. Business Insights & Recommendations

- 1. Revenue concentration:** Electronics is the highest-revenue category in the supplied data. Prioritize stock availability and targeted campaigns for strong categories while investigating underperformers.
- 2. Regional action:** West leads regional revenue. Compare its customer mix, delivery speed and category mix with lower-performing regions before reallocating marketing spend.
- 3. Customer economics:** The customer leaderboard identifies high-value customers. A retention strategy can be built around repeat purchase behavior rather than treating every customer equally.
- 4. Experience matters:** Delivery days and customer ratings can be analyzed together. Regions with slower delivery and weaker ratings are candidates for logistics or service improvements.
- 5. Payment strategy:** Payment-method revenue share helps prioritize checkout reliability, offers and customer support around the most commercially important payment channels.

Category scorecard

product_category	revenue	rating
Electronics	1,829,899.22	2.95
Clothing	1,531,931.72	3.01
Home	982,083.92	2.94
Beauty	765,860.88	3.01

Regional scorecard

region	revenue	delivery
West	1,345,582.16	6.08
North	1,281,508.45	6.12
South	1,246,640.90	6.17
East	1,236,044.23	6.10

Payment mix

payment_method	revenue	orders
Card	2,366,248.20	2270
COD	1,788,408.44	1774
Wallet	955,119.10	956

